

Jonathan Zhu

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SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Backend / Systems: FastAPI, Node.js, Express.js, REST APIs, JWT, OAuth2, Pydantic, SQLAlchemy

Frontend: React, Next.js, TypeScript, Tailwind CSS, Material UI, React Query, Framer Motion

Databases: PostgreSQL, MySQL, MongoDB

Cloud / Infrastructure: AWS (EC2, RDS, S3, Bedrock, Cognito, API Gateway, Amplify, Lambda, Secrets Manager, VPC), Terraform

AI / ML: Amazon Nova Pro, Anthropic API, OpenAI API, Gemini API, Prompt Engineering, Multimodal LLMs

Tools & Compliance: Jenkins, Splunk, Dynatrace, HIPAA Compliance, BAA, pdf2image, asynccpg

WORK EXPERIENCE

IBM <i>Application Developer</i>	Baton Rouge, LA Jun 2026 – Present
Citywide Eye Care <i>Full-Stack Software Engineer</i>	Great Neck, NY Jan 2024 – Jun 2026
- Lead engineer for a production EMR platform used daily by clinical staff, architecting a zero-downtime migration from on-premise servers to AWS (EC2, RDS, S3, Cognito, API Gateway, Amplify) - Owned HIPAA security architecture end to end: Cognito JWT auth, IP whitelisting, role-based access control, and WireGuard VPN, passing all external compliance reviews with zero audit findings - Reduced physician documentation time by 75% by engineering structured charting workflows with auto-save, copy-forward, and S3-backed document storage via pre-signed URLs - Partnered directly with clinicians to define requirements and iterate on features, accelerating feedback cycles and cutting rework across all major releases	
ZLA – AI-Enabled CRM Platform <i>Full-Stack Software Engineer</i>	Rochester, NY Aug 2025 – May 2026
- Built a production CRM from scratch on a 5-person Agile team, delivering lead management, property tracking, campaign workflows, and agent dashboards within tight contract timelines - Engineered an AI-powered lead discovery pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI, cutting manual prospect research time by 60% - Architected a normalized FastAPI and PostgreSQL backend with complex relational schemas; built a React/TypeScript frontend with Kanban pipelines and map-based lead search - Provisioned and managed AWS infrastructure with Terraform (Amplify, EC2 with Elastic IP, API Gateway, Lambda) and built an admin dashboard for one-click backend server management	
Citywide Eye Care <i>Software Engineer Intern</i>	Great Neck, NY Sep 2023 – Dec 2023
- Built and deployed an AI health chatbot using Python and Gemini API, autonomously handling patient FAQs and symptom triage, cutting front-desk call volume by 50% - Rebuilt the patient-facing website in Next.js and Material UI, improving SEO and driving a 25% increase in appointment conversion rates - Implemented digital intake and scheduling workflows on AWS (EC2, S3, RDS, Secrets Manager), replacing all manual paper-based processes	
ADP <i>Site Reliability Engineer Intern</i>	Roseland, NJ Jun 2023 – Aug 2023
- Developed an automated service health and status-reporting tool integrated with Splunk and Dynatrace, eliminating manual monitoring workflows and reducing incident response turnaround time by 20% - Improved deployment reliability and reduced release cycle time by 30% by optimizing Jenkins CI/CD pipelines and tuning MySQL query performance across internal analytics services - Owned observability integrations across S3, Splunk, and Dynatrace to surface system-level metrics, logs, and alerts; recognized by engineering leadership for measurable impact on operational visibility and team velocity	

PROJECTS

- RemitAI:** HIPAA-compliant AI platform for healthcare RCM using Amazon Nova Pro on AWS Bedrock (BAA); processes EOB/ERA PDFs via multimodal pipeline, classifies denials by CARC/RARC codes, generates appeal strategies, and builds a self-learning payer knowledge base on async FastAPI, PostgreSQL, and React/TypeScript
- AI Lead Discovery Pipeline:** Multi-source enrichment pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI to surface qualified real estate prospects, cutting manual research time by 60%
- Enlighten Learning:** Full-stack nonprofit site built in React, TypeScript, and Material UI with serverless API integration (Google Apps Script), dark/light mode, animated UI, and donation tracking; supports tutoring programs for students in underserved communities across Long Island

EDUCATION

Georgia Institute of Technology <i>M.S. in Computer Science</i>	Expected May 2028
Rochester Institute of Technology <i>B.S. in Software Engineering</i>	Expected May 2026